



VLAN Principles and Configuration

CSC 113 – Communication and Networks



Foreword

- Ethernet technology implements data communication over shared media based on carrier sense multiple access with collision detection (CSMA/CD). If there are a large number of PCs on the Ethernet, security risks and broadcast storms may occur, deteriorating network performance and even causing network breakdowns.
- The virtual local area network (VLAN) technology is therefore introduced to solve the preceding problem.
- This course describes basic VLAN principles, working principles of different Layer 2 interfaces, VLAN applications, data forwarding principles, and basic VLAN configuration methods.



Objectives

- On completion of this course, you will be able to:
 - Understand the background of the VLAN technology.
 - Identify the VLAN to which data belongs.
 - Master different VLAN assignment modes.
 - Describe how data communication is implemented through VLANs.
 - Master basic VLAN configuration methods.

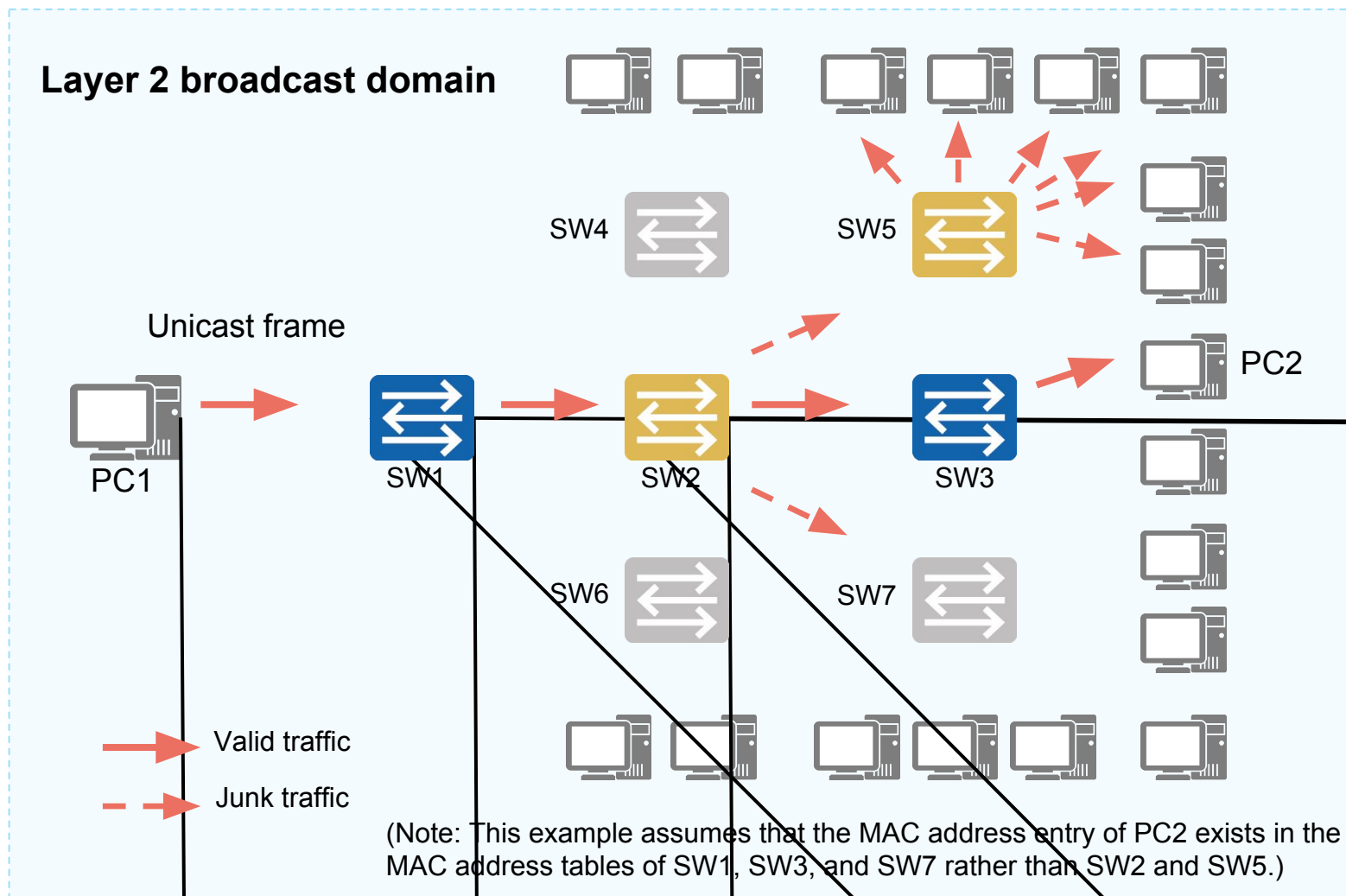


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2. VLAN Principles
3. VLAN Applications
4. VLAN Configuration Examples



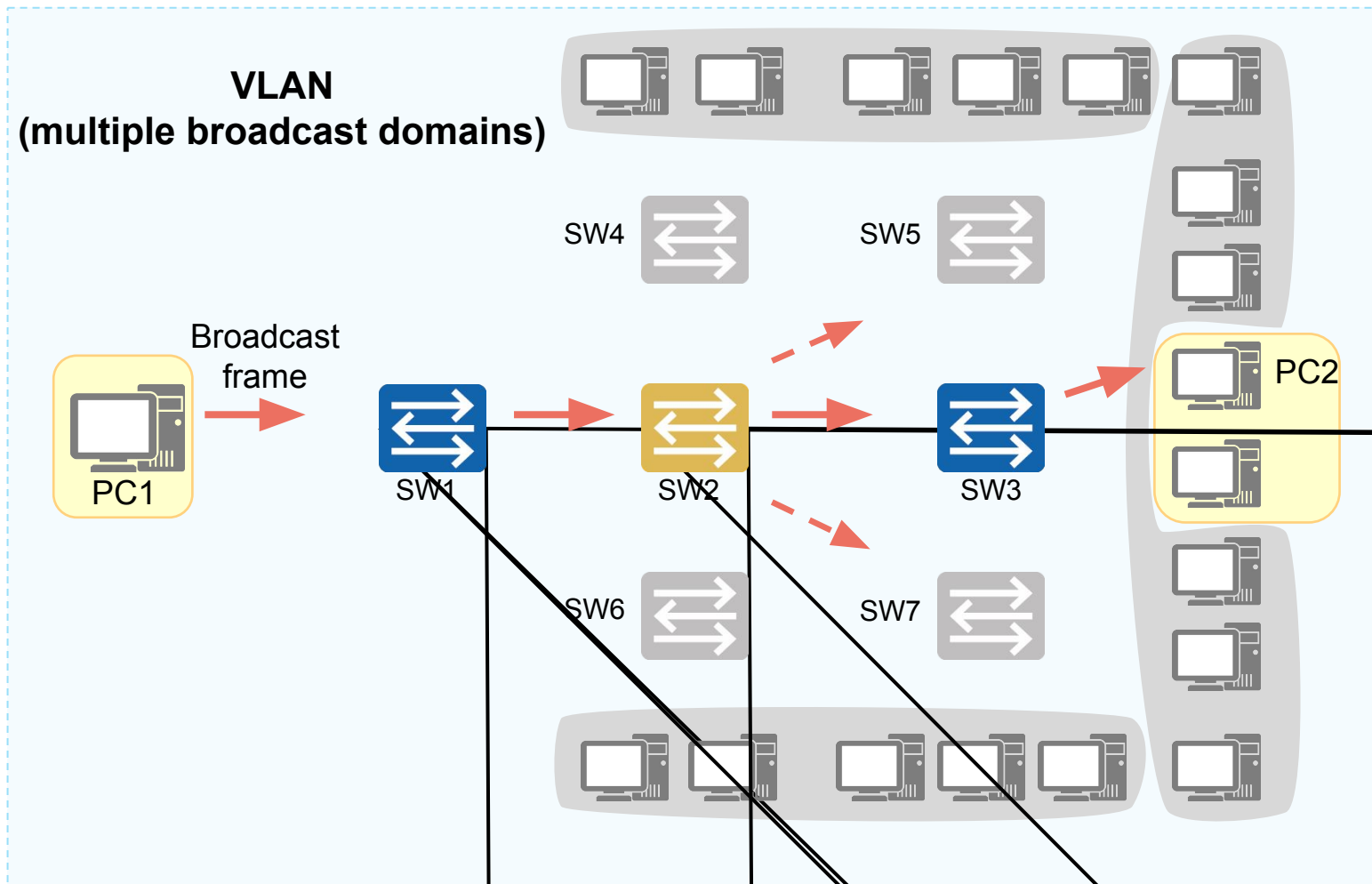
Issues Facing a Traditional Ethernet



- On a typical switching network, broadcast frames or unknown unicast frames sent by a PC are flooded in the entire broadcast domain.
- The larger the broadcast domain is, the more serious network security and junk traffic problems are.



VLAN



- The VLAN technology isolates broadcast domains.
- Characteristics:
 - Geographically independent.
 - Only devices in the same VLAN can directly communicate at Layer 2.

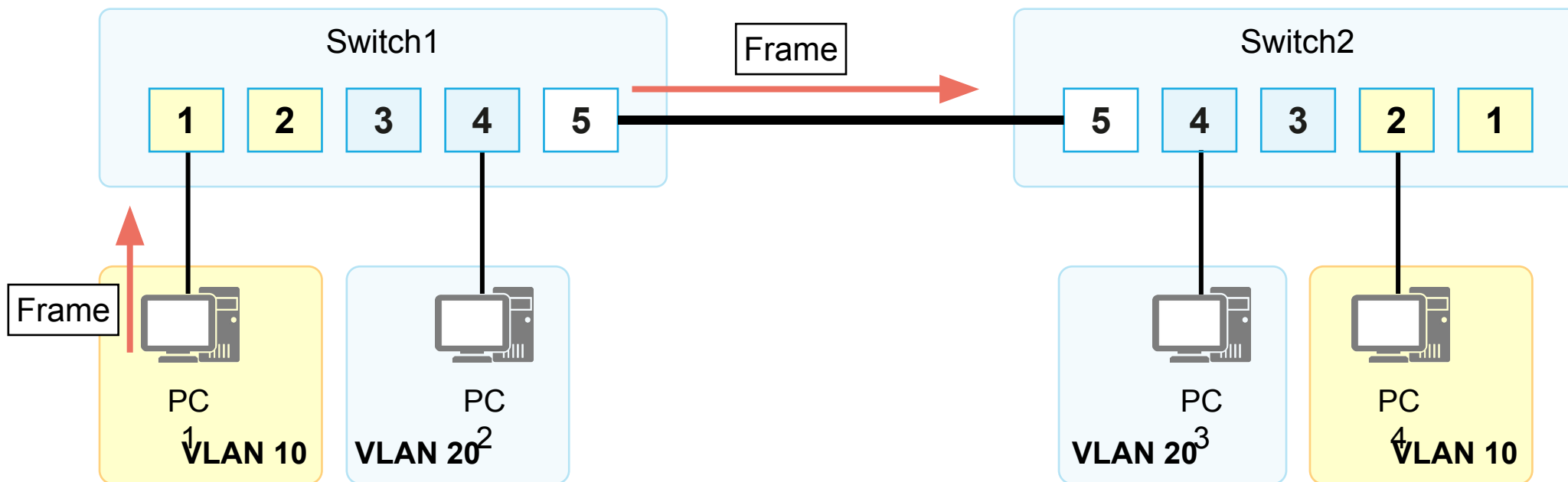


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VLAN Implementation

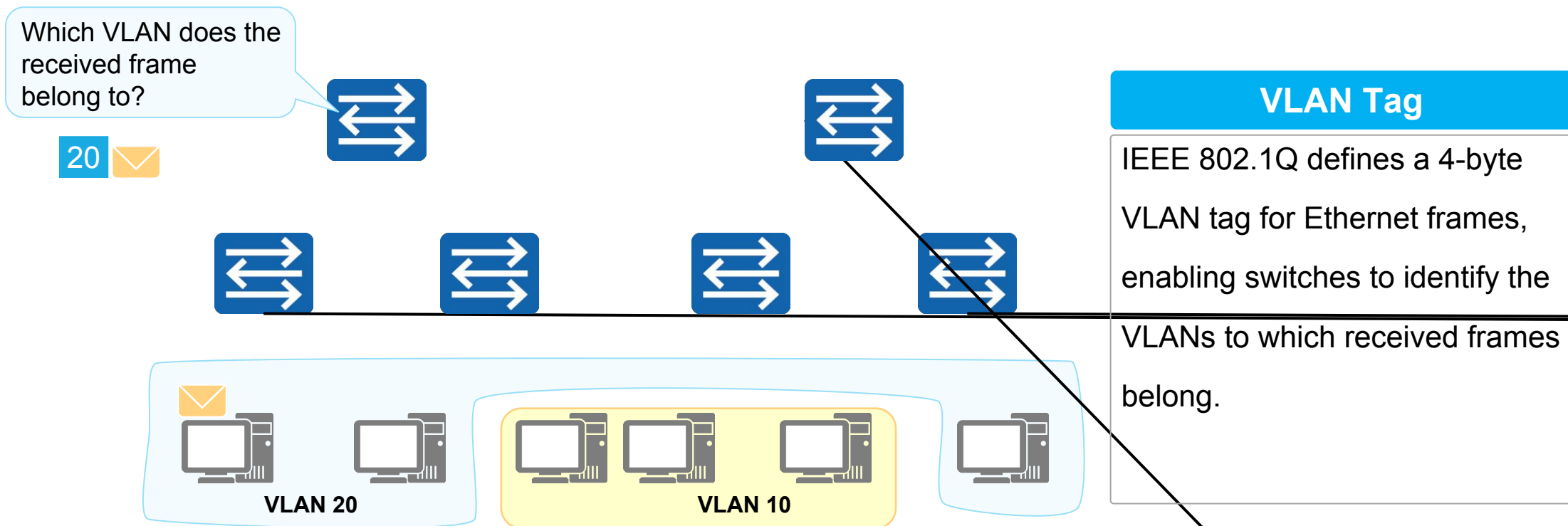


- Switch1 and Switch2 belong to the network of the same enterprise. VLANs are planned for the network, with VLAN 10 for department A and VLAN 20 for department B. Employees in departments A and B are connected to both Switch1 and Switch2.
- Assume that a frame sent from PC1 reaches Switch2 through the link between Switch1 and Switch2. **If no processing is implemented, Switch2 can neither identify the VLAN to which the frame belongs nor determine the local VLAN to which the frame should be sent.**



VLAN Tag

- How does a switch identify the VLAN to which a received frame belongs?





VLAN Frame

Original Ethernet frame
(untagged frame)

Destination MAC address	Source MAC address	Length/ Type	Data	FCS
----------------------------	-----------------------	-----------------	------	-----

802.1Q tag inserted
between the two fields

TPID (0x8100)	PRI	CFI	VLAN ID
16 bits	3 bits	1 bit	12 bits

802.1Q tag

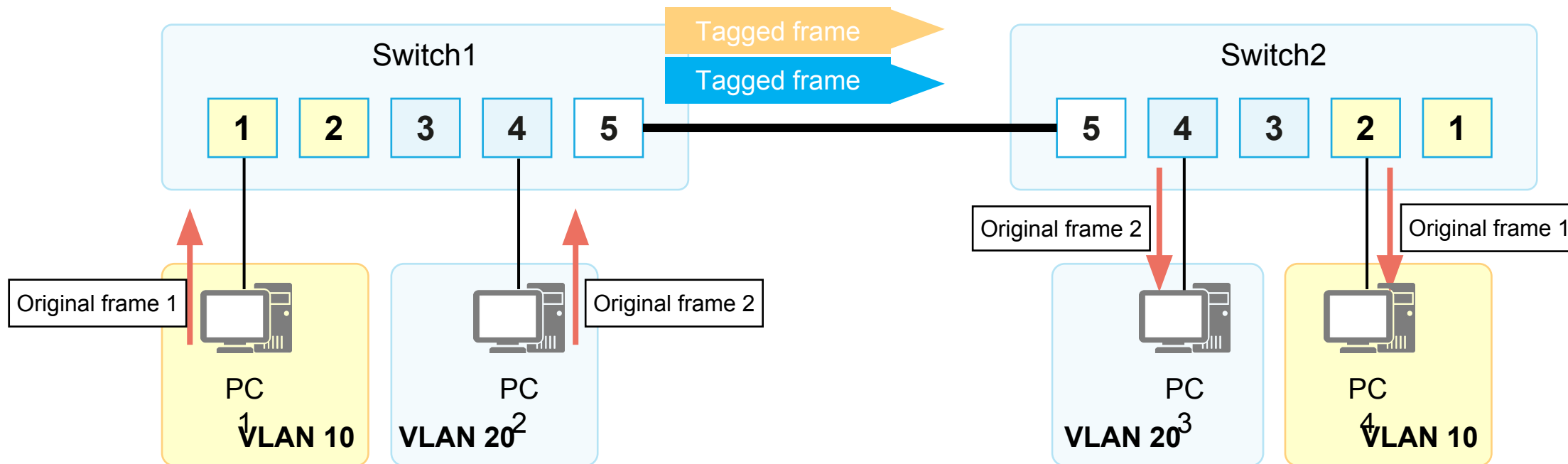
- Tag protocol identifier (TPID): identifies the type of a frame. The value 0x8100 indicates an IEEE 802.1Q frame.
- PRI: identifies the priority of a frame, which is mainly used for QoS.
- Canonical format indicator (CFI): indicates whether a MAC address is in the canonical format. For Ethernet frames, the value of this field is 0.
- VLAN ID: identifies the VLAN to which a frame belongs.

802.1Q frame
(tagged frame)

Destination MAC address	Source MAC address	Tag	Length/ Type	Data	FCS
----------------------------	-----------------------	-----	-----------------	------	-----



VLAN Implementation

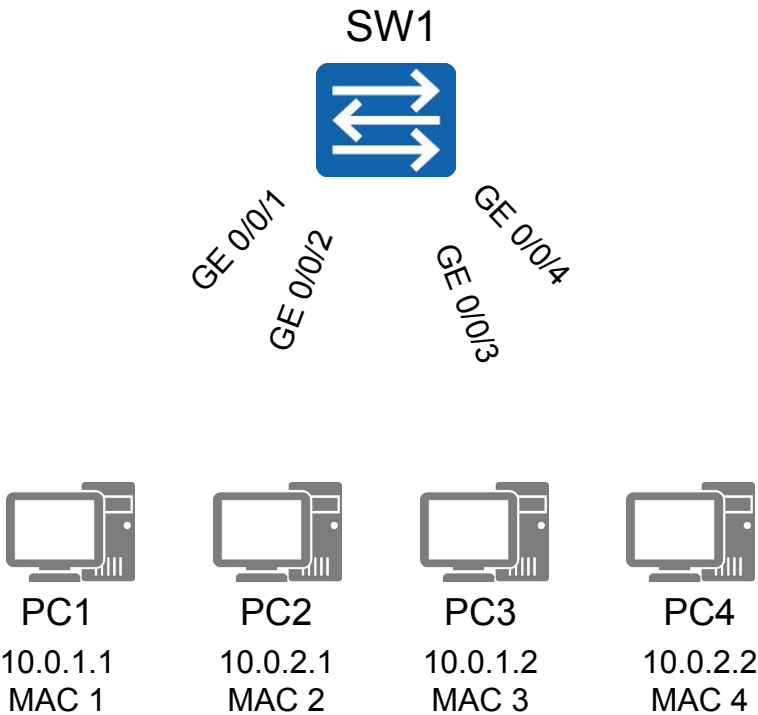


- The link between Switch1 and Switch2 carries data of multiple VLANs. In this situation, a VLAN-based data tagging method is required to distinguish the frames of different VLANs.
- IEEE 802.1Q, often referred to as Dot1q, defines a system of VLAN tagging for Ethernet frames by inserting an 802.1Q tag into the frame header to carry VLAN information.



VLAN Assignment Methods

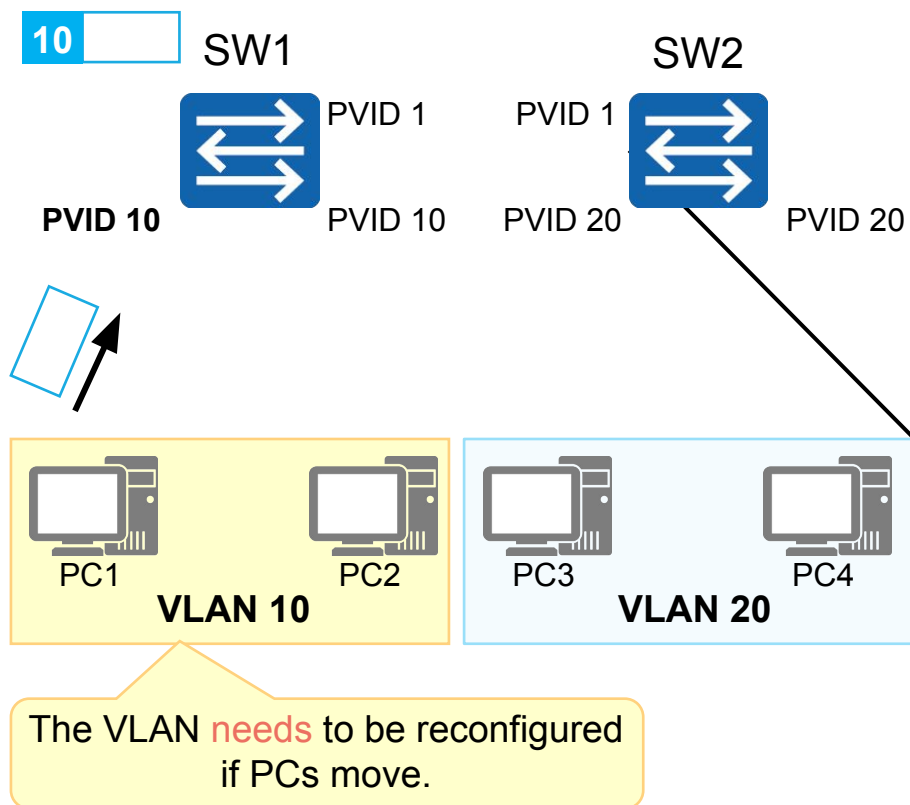
- How are VLANs assigned on a network?



VLAN Assignment Method	VLAN 10	VLAN 20
Interface-based assignment	GE 0/0/1 and GE 0/0/3	GE 0/0/2 and GE 0/0/4
MAC address-based assignment	MAC 1 and MAC 3	MAC 2 and MAC 4
IP subnet-based assignment	10.0.1.*	10.0.2.*
Protocol-based assignment	IP	IPv6
Policy-based assignment	10.0.1.* + GE 0/0/1 + MAC 1	10.0.2.* + GE 0/0/2 + MAC 2



Interface-based VLAN Assignment



Interface-based VLAN Assignment

• Principles

- VLANs are assigned based on interfaces.
- A network administrator preconfigures a PVID for each switch interface and assigns each interface to a VLAN corresponding to the PVID.
- After an interface receives an untagged frame, the switch adds a tag carrying the PVID of the interface to the frame. The frame is then transmitted in the specified VLAN.

• Port Default VLAN ID: PVID

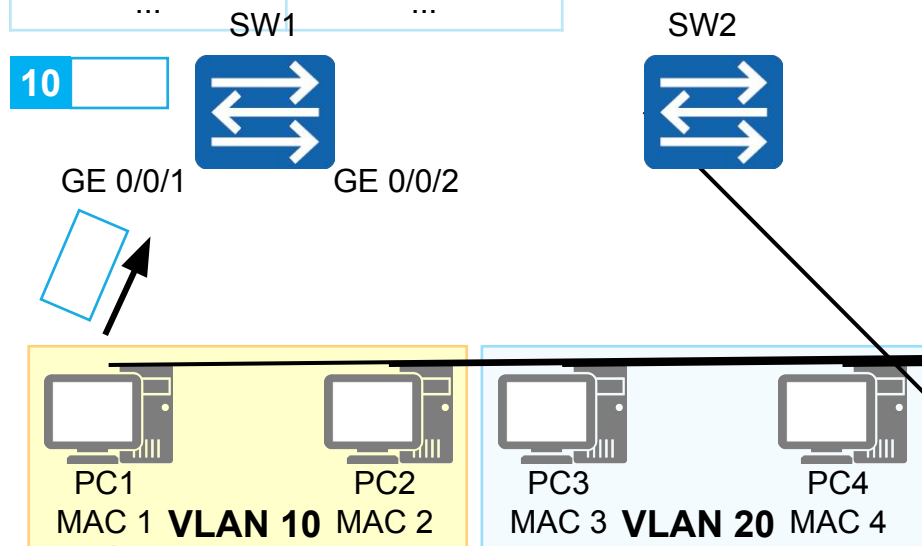
- Default VLAN ID for an interface
- Value range: 1–4094



MAC Address-based VLAN Assignment

Mapping Between MAC Addresses
and VLAN IDs on SW1

MAC Address	VLAN ID
MAC 1	10
MAC 2	10
...	...



MAC Address-based VLAN Assignment

• Principles

- VLANs are assigned based on the source MAC addresses of frames.
- A network administrator preconfigures the mapping between MAC addresses and VLAN IDs.
- After receiving an untagged frame, a switch adds the VLAN tag mapping the source MAC address of the frame to the frame. The frame is then transmitted in the specified VLAN.

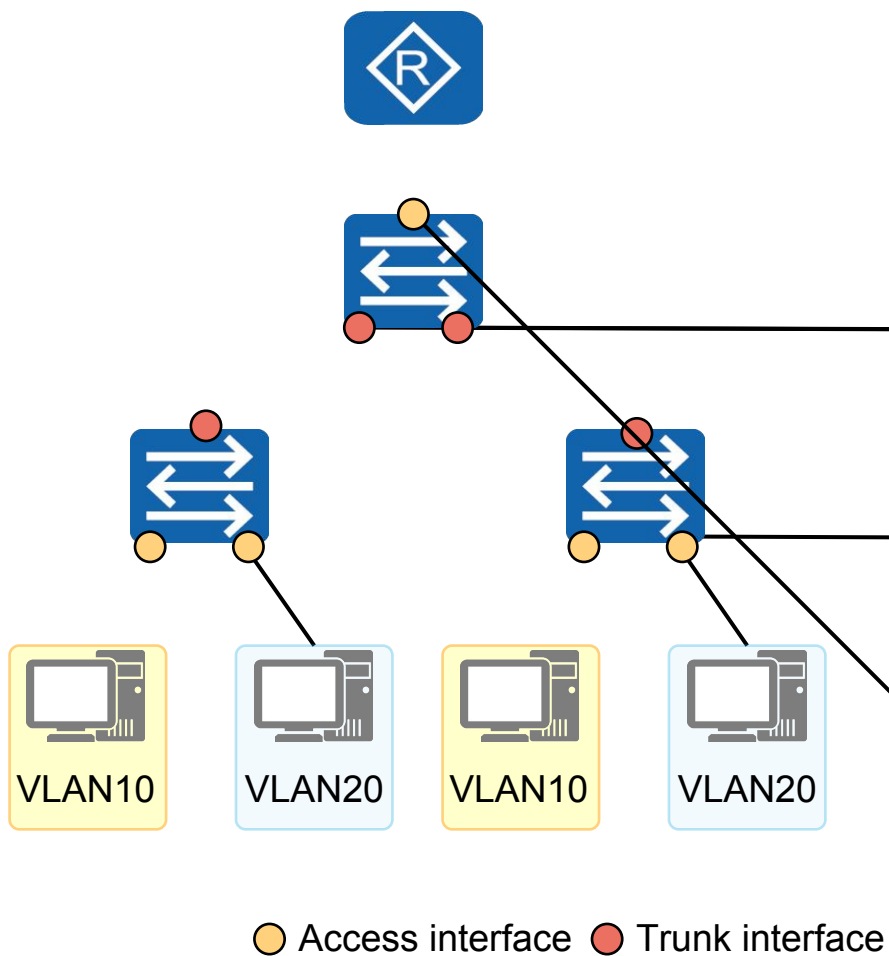
• Mapping table

- Records the mapping between MAC addresses and VLAN IDs.

The VLAN **does not need** to be reconfigured even if PCs move.



Layer 2 Ethernet Interface Types



Interface Types

- **Access interface**

An access interface is used to connect a switch to a terminal, such as a PC or server. In general, the NICs on such a terminal receive and send only untagged frames. An access interface can be added to only one VLAN.

- **Trunk interface**

A trunk interface allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. This type of interface is used to connect a switch to another switch or a sub-interface on a device, such as a router or firewall.

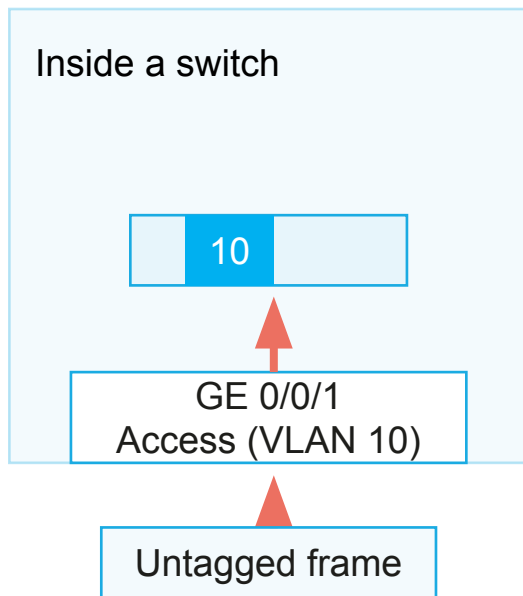
- **Hybrid interface**

Similar to a trunk interface, a hybrid interface also allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. You can determine whether to allow a hybrid interface to carry VLAN tags when sending the frames of one or more VLANs.



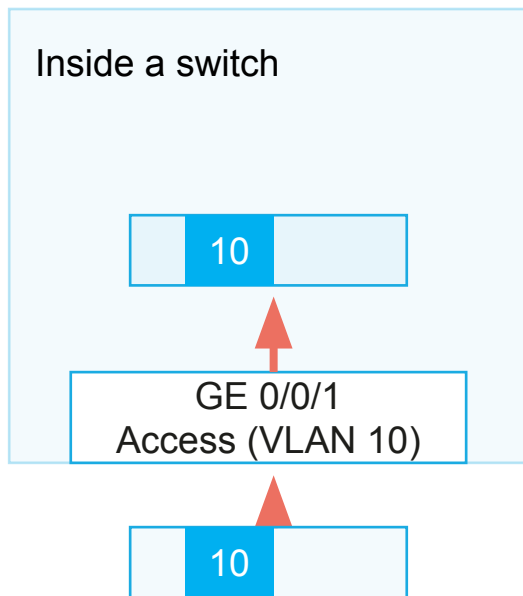
Access Interface

Frame receiving



After receiving an untagged frame:

The interface permits the frame and adds a VLAN tag carrying the PVID of the interface.

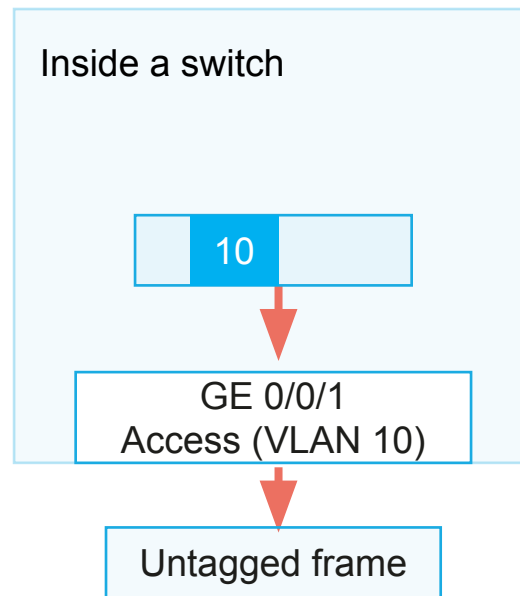


After receiving a tagged frame:

If the VLAN ID of the frame is the same as the PVID of the interface, the interface permits the frame.

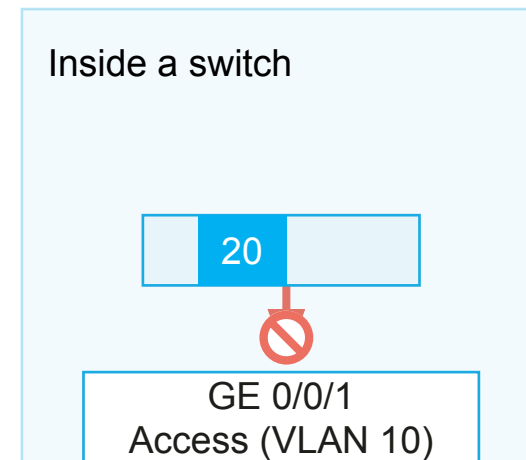
If the VLAN ID of the frame is different from the PVID of the interface, the interface discards the frame.

Frame sending



If the VLAN ID of the frame is the same as the PVID of the interface:

The interface removes the VLAN tag from the frame and then sends the frame.



If the VLAN ID of the frame is different from the PVID of the interface:

The interface discards the frame.

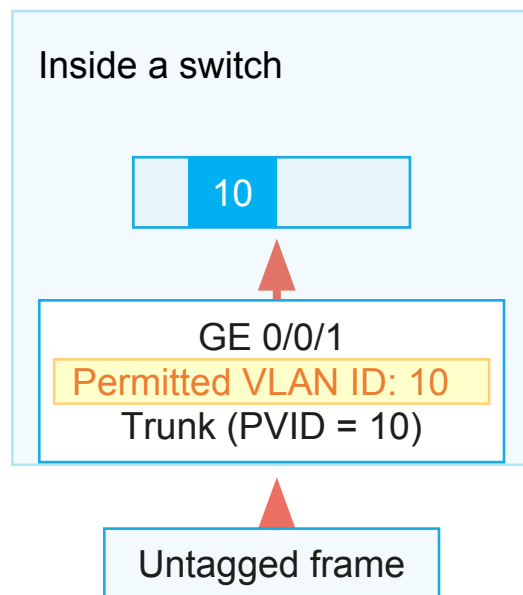
Untagged frame

10 Tagged frame



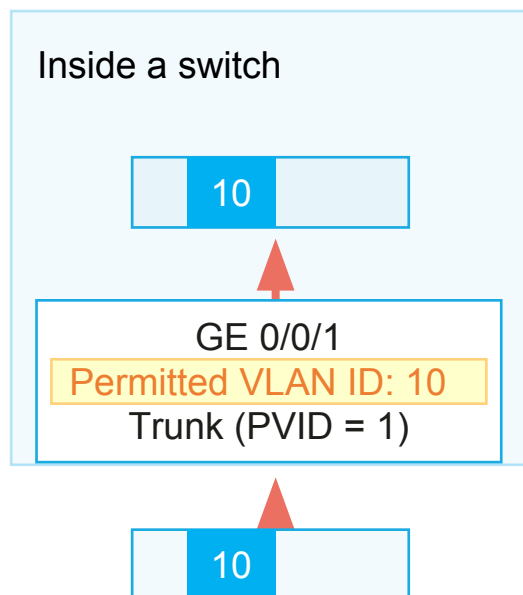
Trunk interface

Frame receiving



After receiving an untagged frame:

The interface adds a VLAN tag with the VID being the PVID of the interface to the frame and permits the frame only when the **VID is in the list of VLAN IDs permitted by the interface**. If the VID is not in the list, the interface discards the frame.



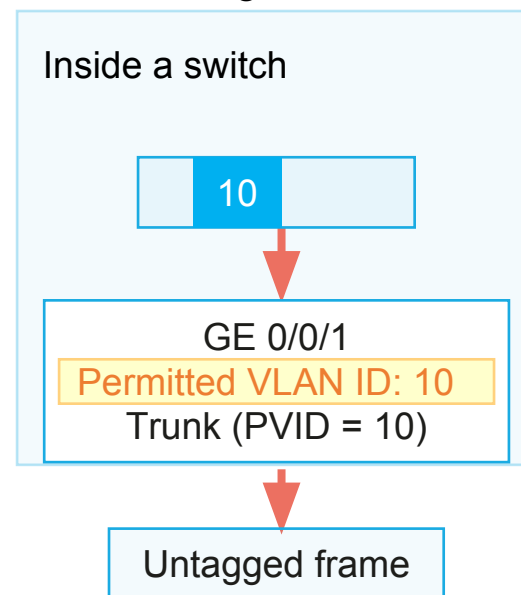
After receiving a tagged frame:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface permits the frame. Otherwise, the interface discards the frame.

Untagged frame

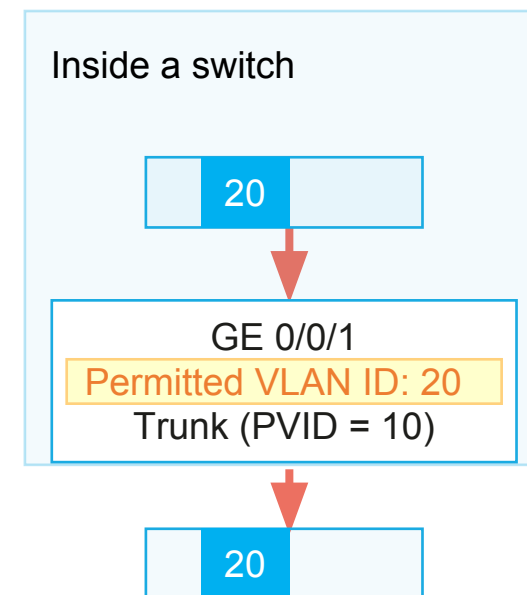
10 Tagged frame

Frame sending



If the VLAN ID of the frame is the same as the PVID of the interface:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface removes the tag from the frame and sends the frame out. Otherwise, the interface discards the frame.



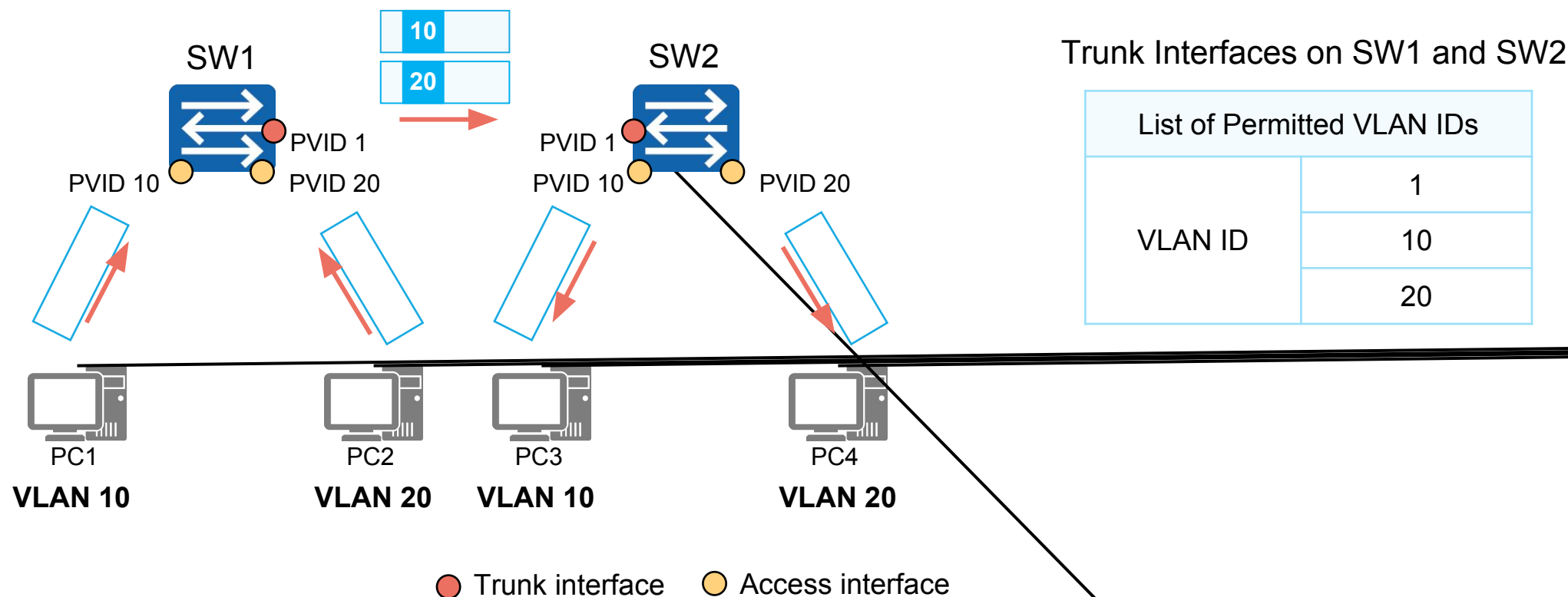
If the VLAN ID of the frame is different from the PVID of the interface:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface sends the frame out without removing the tag of the frame. Otherwise, the interface discards the frame.



Example for Frame Processing on Access and Trunk Interfaces

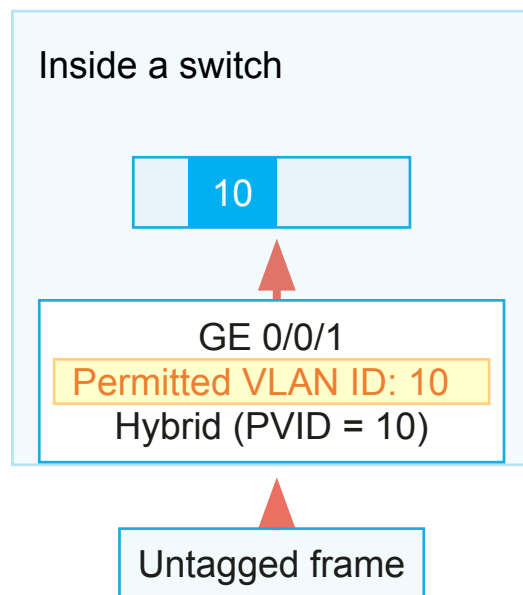
- Describe how inter-PC access is implemented in this example.





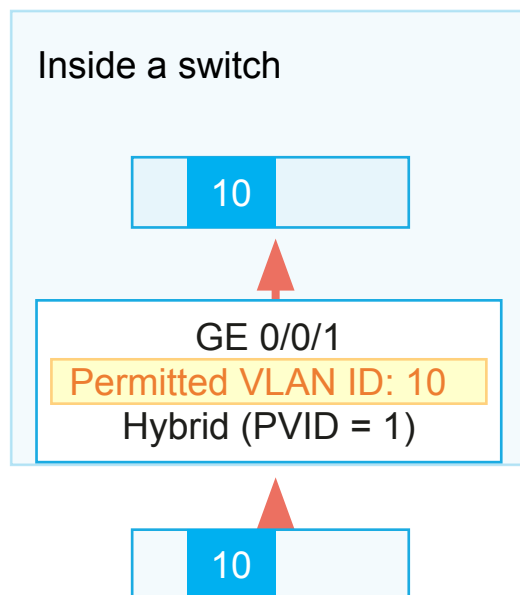
Hybrid Interface

Frame receiving



After receiving an untagged frame:

The interface adds a VLAN tag with the VID being the PVID of the interface to the frame and permits the frame only when the VID is in the list of VLAN IDs permitted by the interface. If the VID is not in the list, the interface discards the frame.

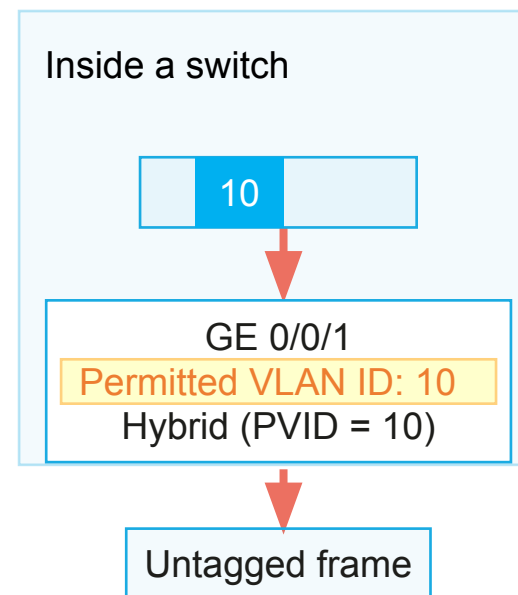


After receiving a tagged frame:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface permits the frame. Otherwise, the interface discards the frame.

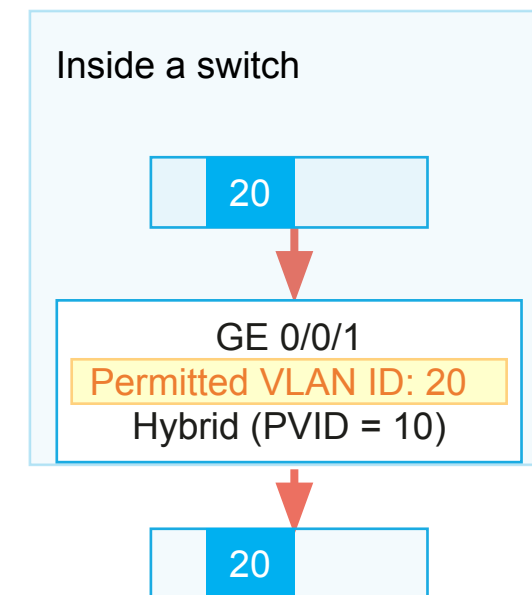


Frame sending



If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface:

If the interface has been configured not to carry VLAN tags when sending frames, it removes the tag from the frame and then sends the frame out.



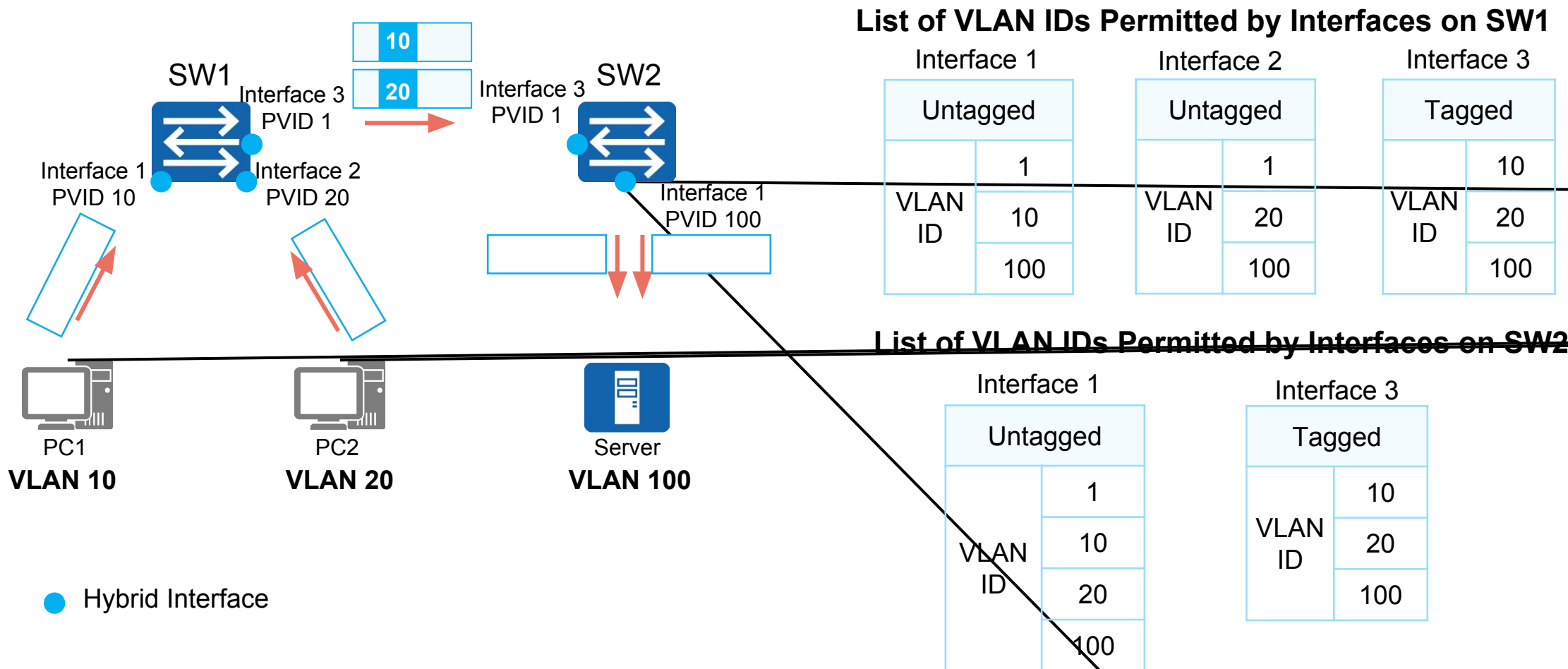
If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface:

If the interface has been configured to carry VLAN tags when sending frames, it sends the frame out without removing the tag of the frame.



Example for Frame Processing on Hybrid Interfaces

- Describe how PCs access the server in this example.





Summary

Access Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and permits the frame.
- Tagged frame: checks whether the VID in the tag of the frame is the same as the PVID of the interface. If they are the same, permits the frame; otherwise, discards the frame.

Frame sending

- Checks whether the VID in the tag of the frame is the same as the PVID of the interface. If they are the same, removes the tag and sends the frame out; otherwise, discards the frame.

Trunk Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.
- Tagged frame: checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.

Frame sending

- If the VID is in the list of permitted VLAN IDs and the same as the PVID of the interface, removes the tag and sends the frame out.
- If the VID is in the list of permitted VLAN IDs but different from the PVID of the interface, sends the frame out without removing the tag.
- If the VID is not in the list of permitted VLAN IDs, discards the frame.

Hybrid Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.
- Tagged frame: checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.

Frame sending

- If the VID is not in the list of permitted VLAN IDs, discards the frame.
- If the VID is in the untagged VLAN ID list, removes the tag and sends the frame out.
- If the VID is in the tagged VLAN ID list, sends the frame out without removing the tag.



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VLAN Planning

- **VLAN assignment rules**

- By service: voice, video, and data VLANs
- By department: e.g. VLANs for engineering, marketing, and financing departments
- By application: e.g. VLANs for servers, offices, and classrooms

- **Example for VLAN planning**

- Assume that there are three buildings: administrative building with offices, classrooms, and financing sections, teaching building with offices and classrooms, and office building with offices and financing sections. Each building has one access switch, and the core switch is deployed in the administrative building.
- The following table describes the VLAN plan.

VLAN ID	IP Address Segment	Description
1	X.16.10.0/24	VLAN to which office users belong
2	X.16.20.0/24	VLAN to which the users of the financing department belong
3	X.16.30.0/24	VLAN to which classroom users belong
100	Y.16.100.0/24	VLAN to which the device management function belongs

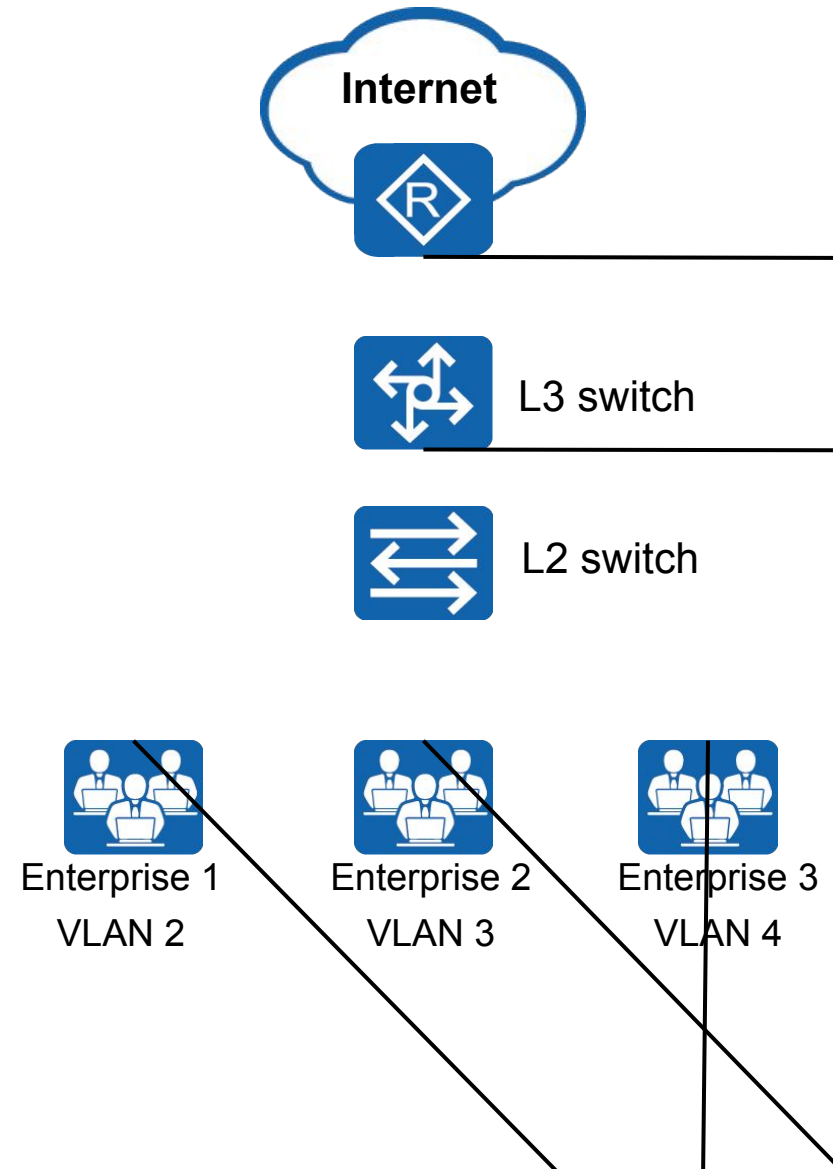
- **Tips for VLAN assignment**

VLAN IDs can be randomly assigned within the supported range. To improve VLAN ID continuity, you can associate VLAN IDs with subnets during VLAN assignment.



Interface-based VLAN Assignment

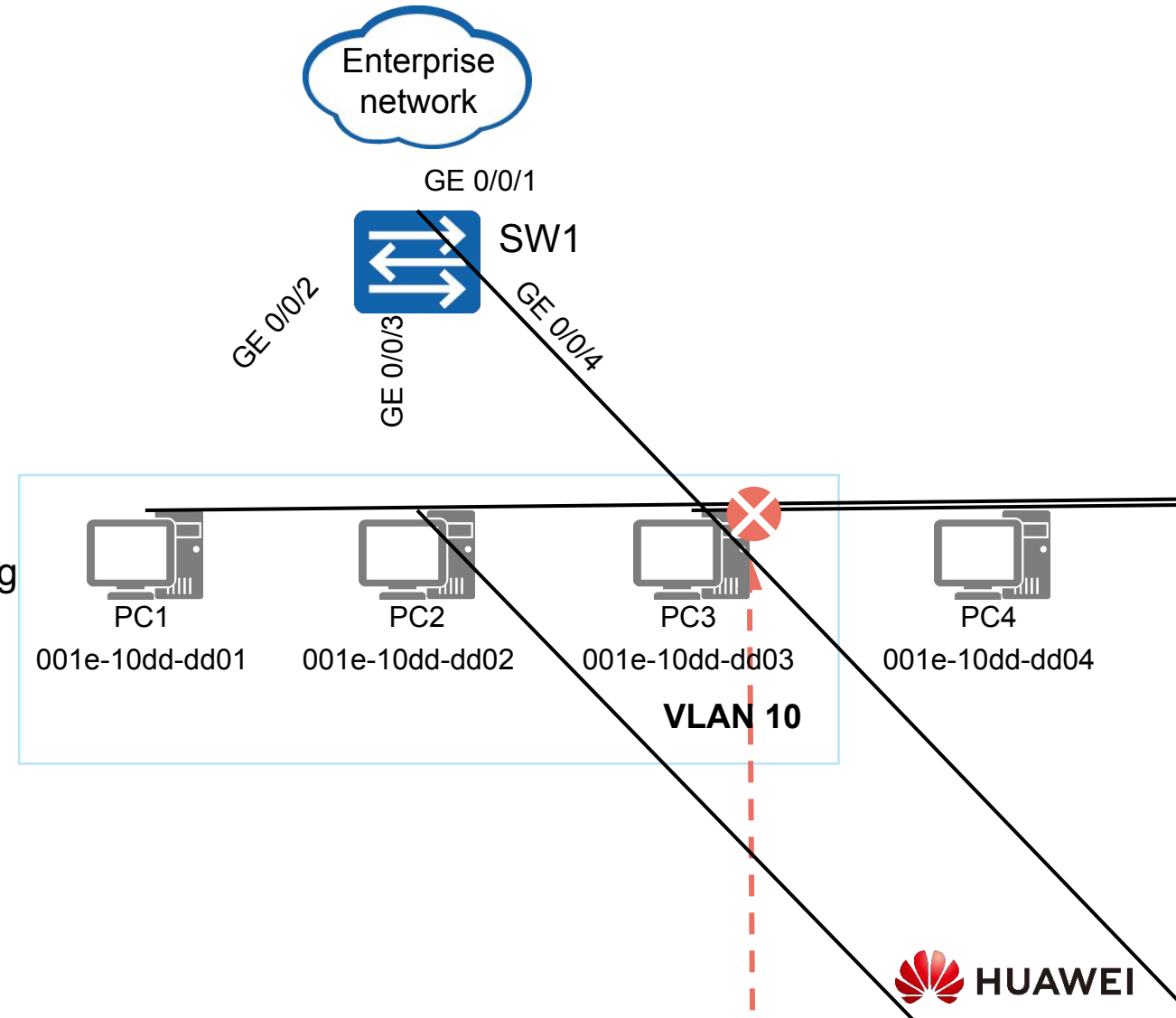
- Applicable scenario:
 - There are multiple enterprises in a building. These enterprises share network resources to reduce costs. Networks of the enterprises connect to different interfaces of the same Layer 2 switch and access the Internet through the same egress device.
- VLAN assignment:
 - To isolate the services of different enterprises and ensure service security, assign interfaces connected to the enterprises' networks to different VLANs. In this way, each enterprise has an independent network, and each VLAN works as a virtual work group.





MAC Address-based VLAN Assignment

- Applicable scenario:
 - The network administrator of an enterprise assigns PCs in the same department to the same VLAN. To improve information security, the enterprise requires that only employees in the specified department be allowed to access specific network resources.
- VLAN assignment:
 - To meet the preceding requirement, configure MAC address-based VLAN assignment on SW1, preventing new PCs connected to the network from accessing the network resources.





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Basic VLAN Configuration Commands

1. Create one or more VLANs.

```
[Huawei] vlan vlan-id
```

This command creates a VLAN and displays the VLAN view. If the VLAN to be created already exists, this command directly displays the VLAN view.

- The value of *vlan-id* is an integer ranging from 1 to 4094.

```
[Huawei] vlan batch { vlan-id1 [ to vlan-id2 ] }
```

This command creates VLANs in a batch. In this command:

- **batch**: creates VLANs in a batch.
- *vlan-id1*: specifies a start VLAN ID.
- *vlan-id2*: specifies an end VLAN ID.



Basic Access Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type access
```

In the interface view, set the link type of the interface to access.

2. Configure a default VLAN for the access interface.

```
[Huawei-GigabitEthernet0/0/1] port default vlan vlan-id
```

In the interface view, configure a default VLAN for the interface and add the interface to the VLAN.

- *vlan-id*: specifies an ID for the default VLAN. The value is an integer ranging from 1 to 4094.



Basic Trunk Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type trunk
```

In the interface view, set the link type of the interface to trunk.

2. Add the trunk interface to specified VLANs.

```
[Huawei-GigabitEthernet0/0/1] port trunk allow-pass vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the trunk interface to specified VLANs.

3. (Optional) Configure a default VLAN for the trunk interface.

```
[Huawei-GigabitEthernet0/0/1] port trunk pvid vlan vlan-id
```

In the interface view, configure a default VLAN for the trunk interface.



Basic Hybrid Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type hybrid
```

In the interface view, set the link type of the interface to hybrid.

2. Add the hybrid interface to specified VLANs.

```
[Huawei-GigabitEthernet0/0/1] port hybrid untagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs in untagged mode.

```
[Huawei-GigabitEthernet0/0/1] port hybrid tagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs in tagged mode.

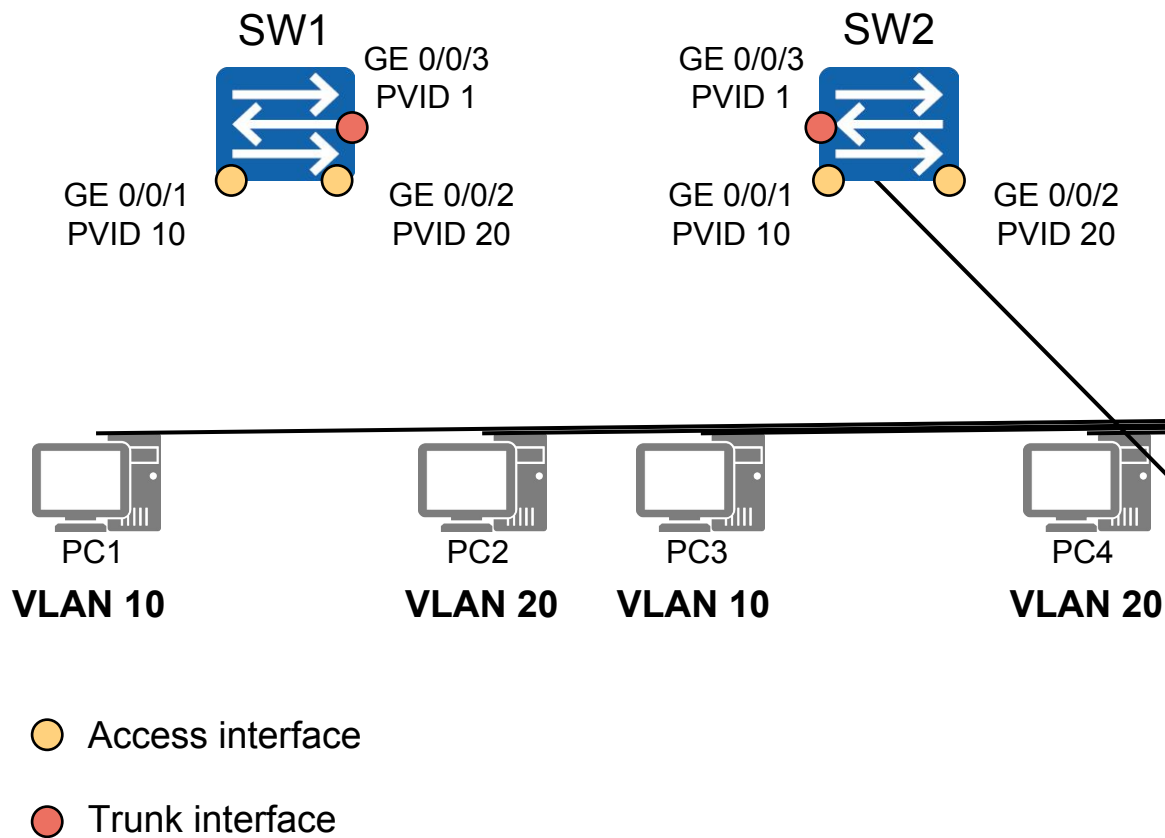
3. (Optional) Configure a default VLAN for the hybrid interface.

```
[Huawei-GigabitEthernet0/0/1] port hybrid pvid vlan vlan-id
```

In the interface view, configure a default VLAN for the hybrid interface.



Case1:Configuring Interface-based VLAN Assignment

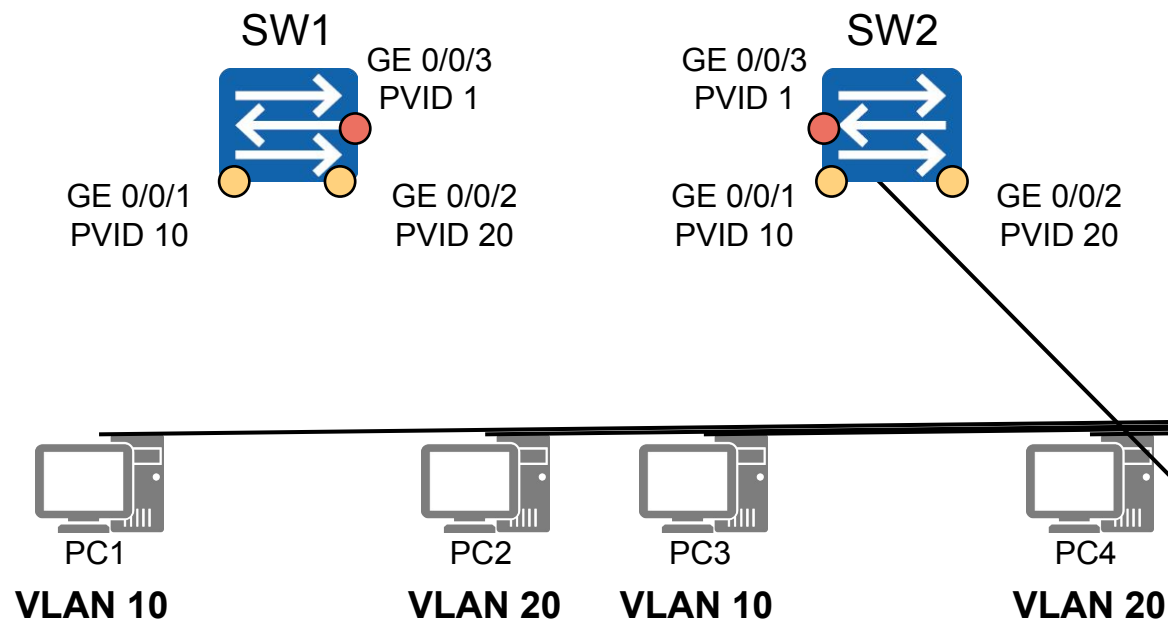


- Networking requirements:

- On the network shown in the left figure, the switches (SW1 and SW2) of an enterprise are connected to multiple PCs, and PCs with the same services access the network using different devices. To ensure communication security, the enterprise requires that only PCs with the same service can directly communicate.
- To meet this requirement, configure interface-based VLAN assignment on the switches and add interfaces connected to PCs with the same service to the same VLAN. In this way, PCs in different VLANs cannot directly communicate at Layer 2, but PCs in the same VLAN can directly communicate.



Creating VLANs



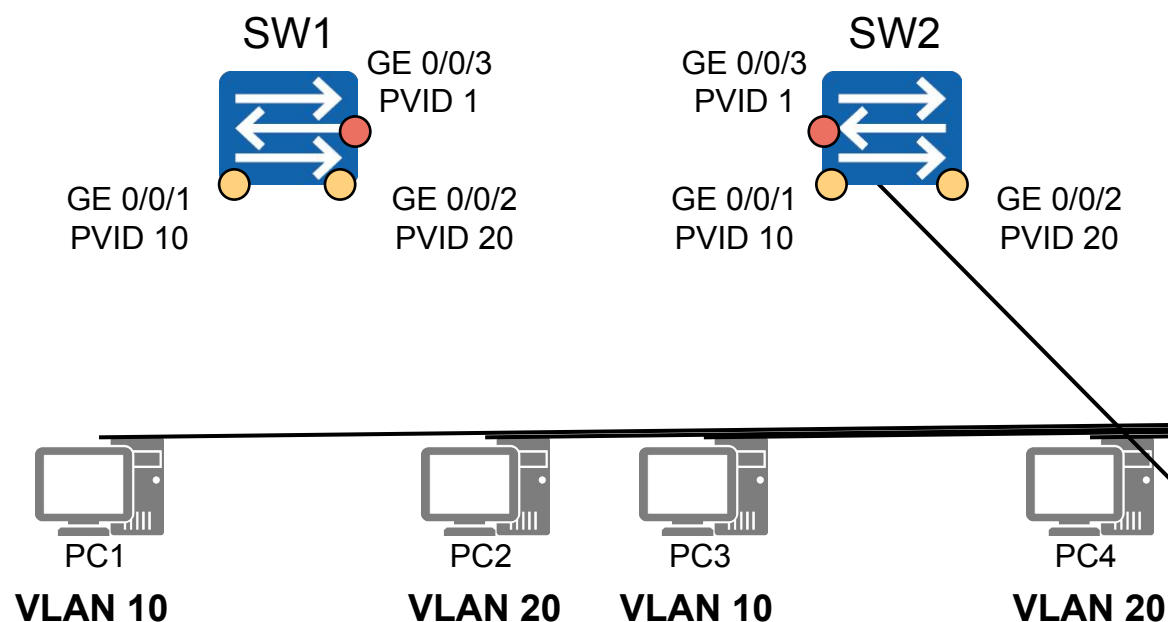
Create VLANs.

```
[SW1] vlan 10
[SW1-vlan10] quit
[SW1] vlan 20
[SW1-vlan20] quit
```

```
[SW2] vlan batch 10 20
```



Configuring Access and Trunk Interfaces



Configure access interfaces and add the interfaces to corresponding VLANs.

```
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type access
[SW1-GigabitEthernet0/0/1] port default vlan 10
```

```
[SW1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type access
[SW1] vlan 20
[SW1-vlan20] port GigabitEthernet0/0/2
[SW1-vlan20] quit
```

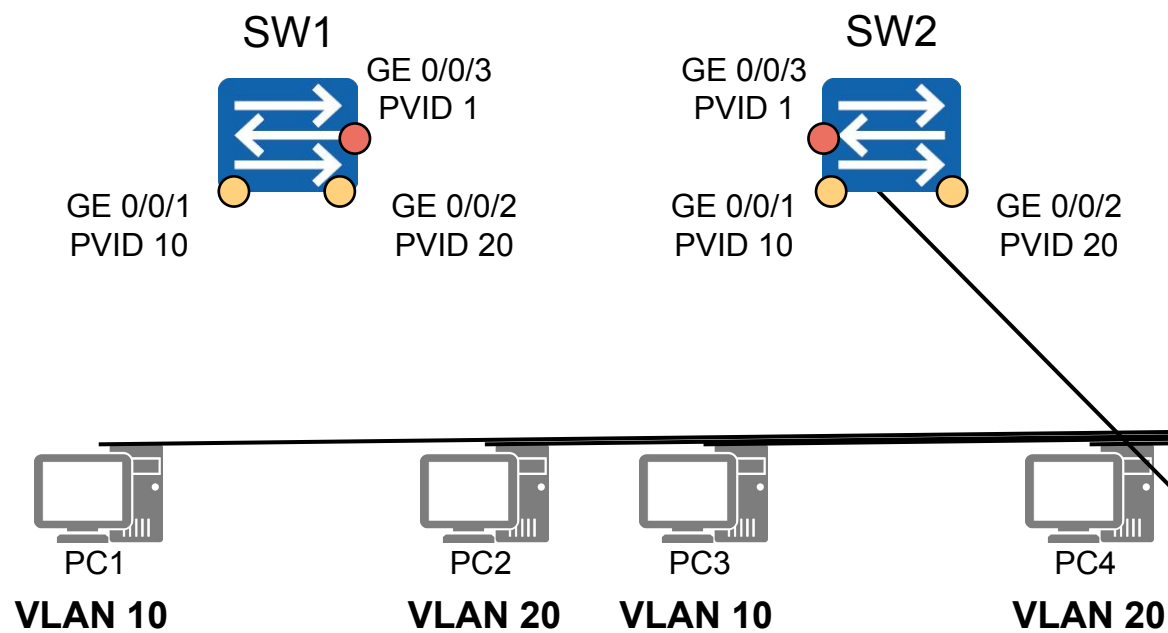
Configure a trunk interface and specify a list of VLAN IDs permitted by the interface.

```
[SW1] interface GigabitEthernet 0/0/3
[SW1-GigabitEthernet0/0/3] port link-type trunk
[SW1-GigabitEthernet0/0/3] port trunk pvid vlan 1
[SW1-GigabitEthernet0/0/3] port trunk allow-pass vlan 10 20
```

Note: The configuration on SW2 is similar to that on SW1.



Verifying the Configuration



```
[SW1]display vlan
```

The total number of vlans is : 3

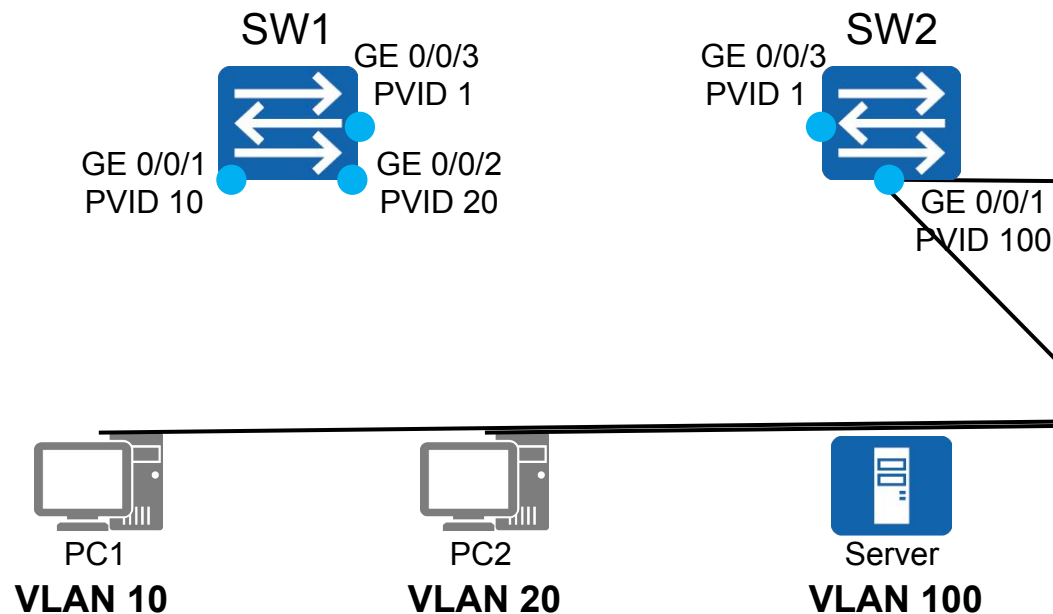
U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
-----	------	-------

1	common	UT:GE0/0/3(U)
10	common	UT:GE0/0/1(U) TG:GE0/0/3(U)
20	common	UT:GE0/0/2(U) TG:GE0/0/3(U)
.....		



Case2:Configuring Interface-based VLAN Assignment



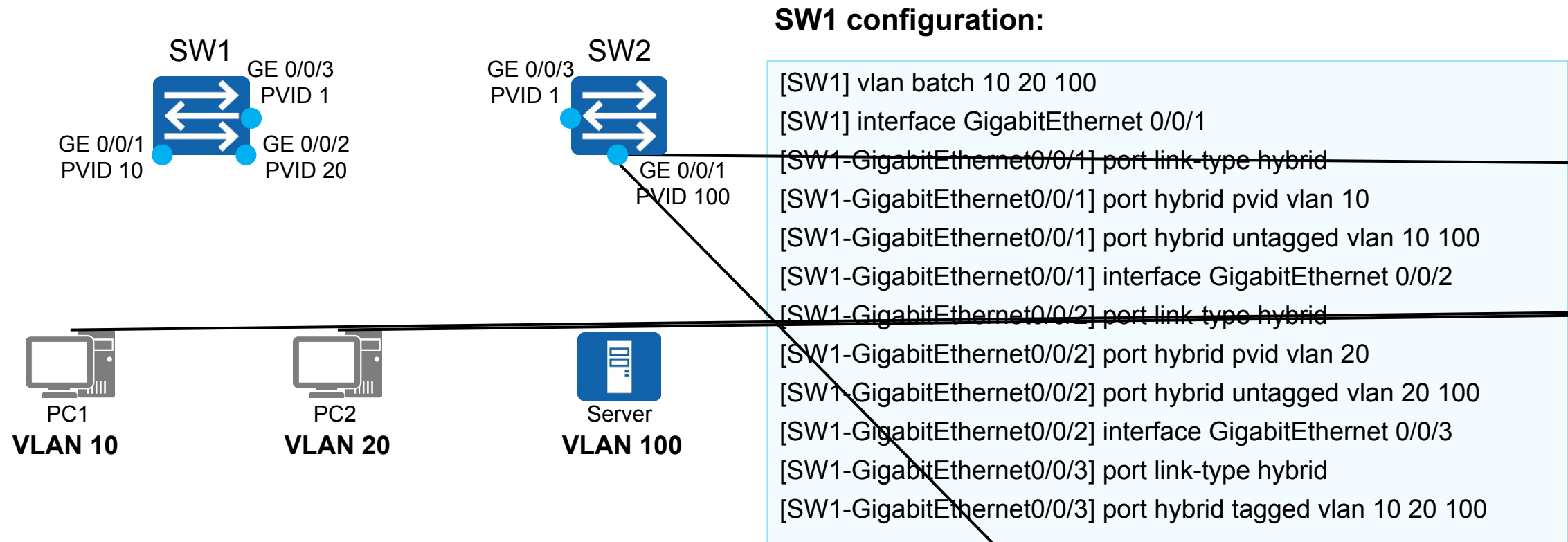
● Hybrid interface

- Networking requirements:

- On the network shown in the left figure, the switches (SW1 and SW2) of an enterprise are connected to multiple PCs, and PCs in different departments need to access the server of the enterprise. To ensure communication security, the enterprise requires that PCs in different departments cannot directly communicate.
- To meet this requirement, configure interface-based VLAN assignment and hybrid interfaces on the switches to enable PCs in different departments to access the server but disable them from directly communicating at Layer 2.

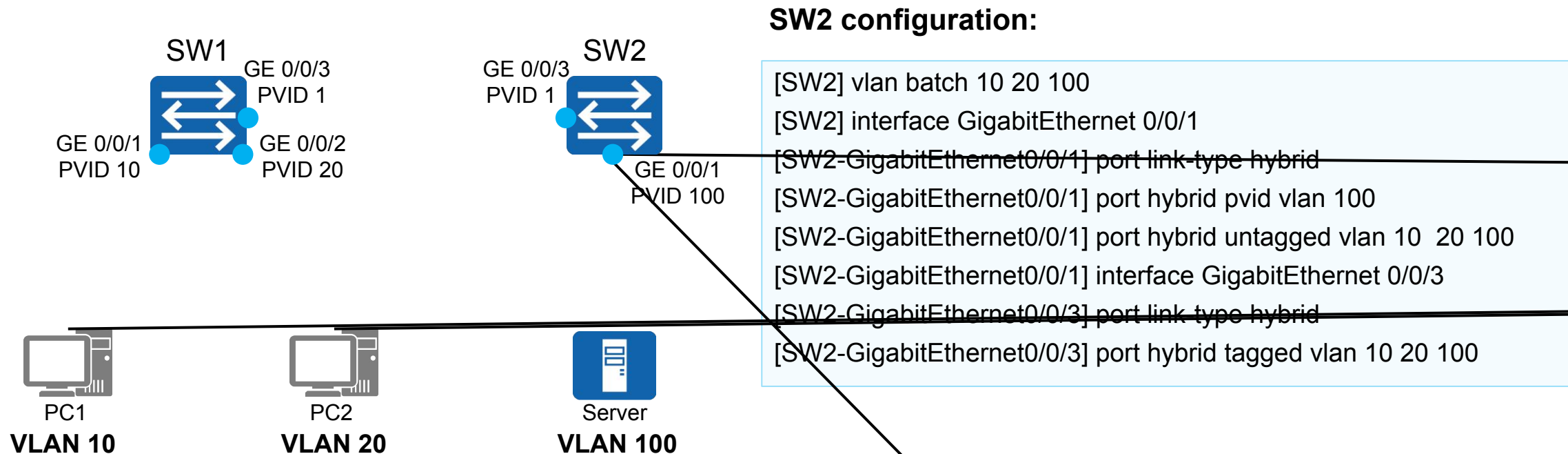


Configuring Hybrid Interfaces (1)



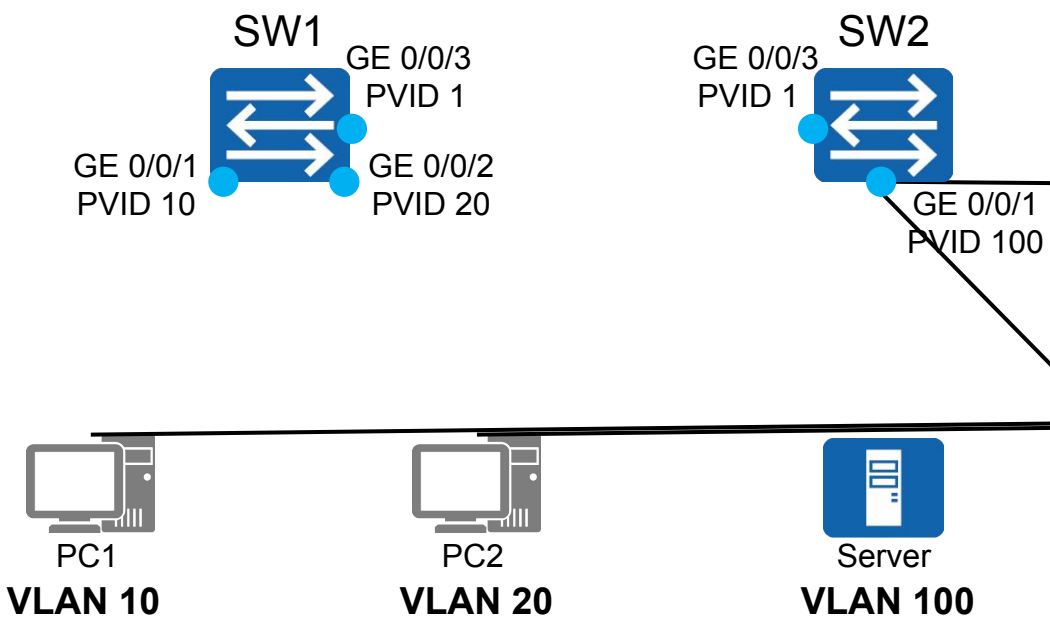


Configuring Hybrid Interfaces (2)





Verifying the Configuration



```
[SW1]display vlan
The total number of vlans is : 4
-----
U: Up; D: Down;          TG: Tagged; UT: Untagged;
MP: Vlan-mapping;        ST: Vlan-stacking;
#. ProtocolTransparent-vlan; *: Management-vlan;
-----
VID   Type   Ports
-----
1     common  UT:GE0/0/1(U)   GE0/0/2(U)   GE0/0/3(U) .....
10    common  UT:GE0/0/1(U)   TG:GE0/0/3(U)
20    common  UT:GE0/0/2(U)   TG:GE0/0/3(U)
100   common  UT:GE0/0/1(U)   GE0/0/2(U)   TG:GE0/0/3(U)
.....
```



Basic VLAN Configuration Commands

1. Associate a MAC address with a VLAN.

```
[Huawei-vlan10] mac-vlan mac-address mac-address [ mac-address-mask | mac-address-mask-length ]
```

This command associates a MAC address with a VLAN.

- *mac-address*: specifies the MAC address to be associated with a VLAN. The value is a hexadecimal number in the format of H-H-H. Each H contains one to four digits, such as 00e0 or fc01. If an H contains less than four digits, the left-most digits are padded with zeros. For example, e0 is displayed as 00e0. The MAC address cannot be 0000-0000-0000, FFFF-FFFF-FFFF, or any multicast address.
- *mac-address-mask*: specifies the mask of a MAC address. The value is a hexadecimal number in the format of H-H-H. Each H contains one to four digits.
- *mac-address-mask-length*: specifies the mask length of a MAC address. The value is an integer ranging from 1

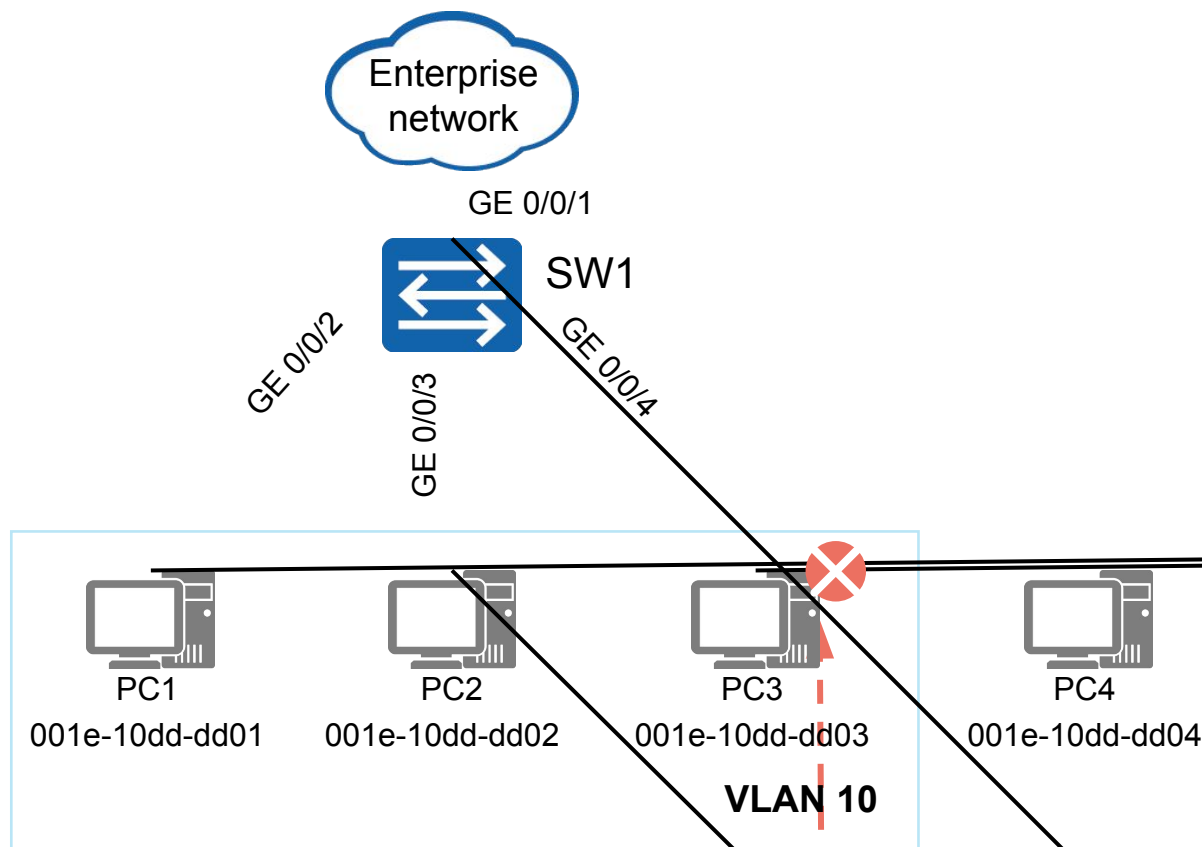
2. Enable MAC address-based VLAN assignment on an interface.

```
[Huawei-GigabitEthernet0/0/1] mac-vlan enable
```

This command enables MAC address-based VLAN assignment on an interface.



Example for Configuring MAC Address-based VLAN Assignment

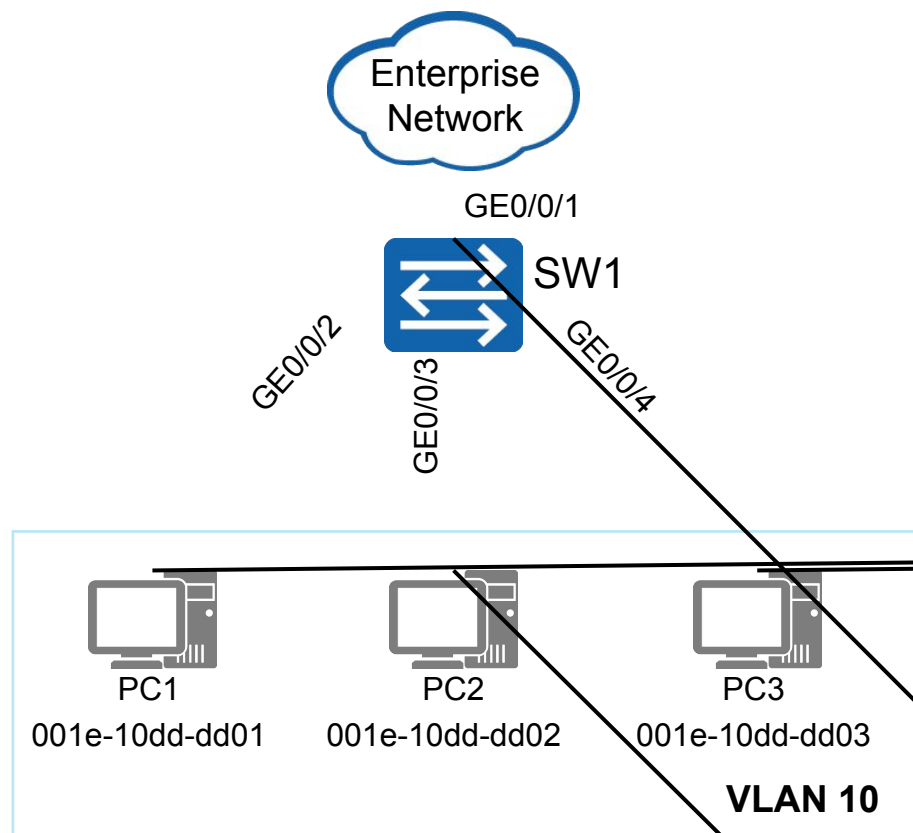


- Networking requirements:

- The network administrator of an enterprise assigns PCs in the same department to the same VLAN. To improve information security, the enterprise requires that only employees in the department be allowed to access the network resources of the enterprise.
- PCs 1 through 3 belong to the same department. According to the enterprise' requirement, only the three PCs can access the enterprise network through SW1.
- To meet this requirement, configure MAC address-based VLAN assignment and associate the MAC addresses of the three PCs with the specified VLAN.



Creating a VLAN and Associating MAC Addresses with the VLAN



Create a VLAN.

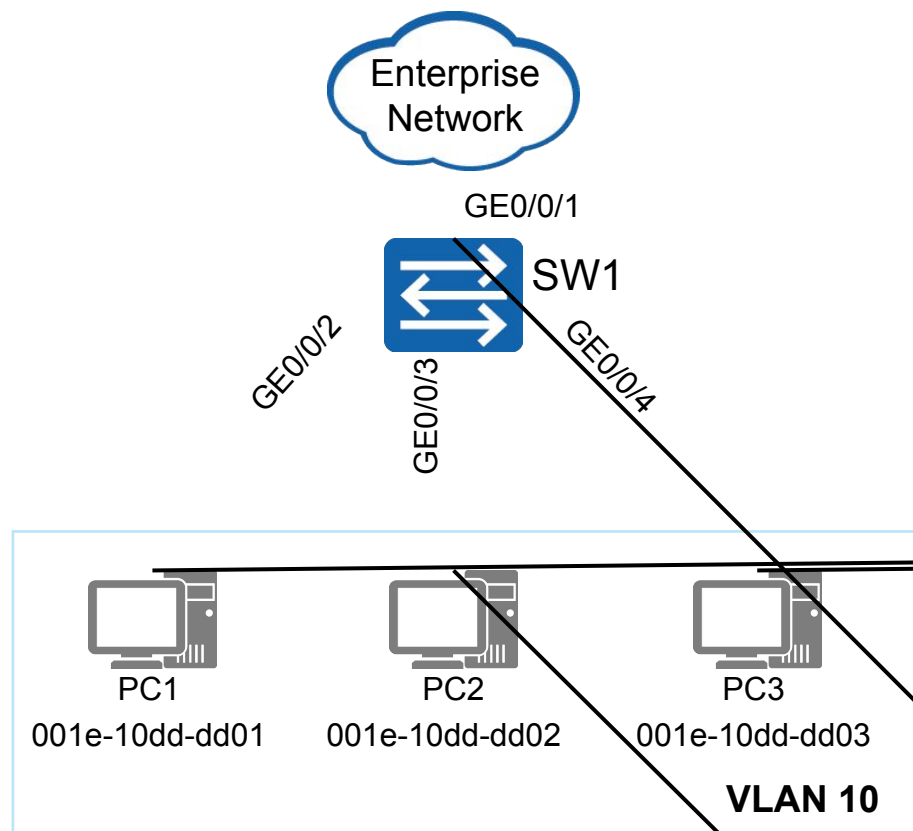
```
[SW1] vlan 10  
[SW1-vlan10] quit
```

Associate MAC addresses with the VLAN.

```
[SW1] vlan 10  
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd01  
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd02  
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd03  
[SW1-vlan10] quit
```



Adding Interfaces to the VLAN and Enabling MAC Address-based VLAN Assignment



Add interfaces to the VLAN.

```
[SW1] interface gigabitethernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type hybrid
[SW1-GigabitEthernet0/0/1] port hybrid tagged vlan 10
```

```
[SW1] interface gigabitethernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type hybrid
[SW1-GigabitEthernet0/0/2] port hybrid untagged vlan 10
```

Enable MAC address-based VLAN assignment on the specified interface.

```
[SW1] interface gigabitethernet 0/0/2
[SW1-GigabitEthernet0/0/2] mac-vlan enable
[SW1-GigabitEthernet0/0/2] quit
```

Note: The configuration of GE 0/0/3 and GE 0/0/4 is similar to that of GE 0/0/2.



Verifying the Configuration

```
[SW1]display vlan
```

The total number of vlans is : 2

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
-----	------	-------

1	common	UT:GE0/0/1(U) GE0/0/2(U) GE0/0/3(U)
---	--------	---

10	common	UT:GE0/0/2(U) GE0/0/3(U) GE0/0/4(U)
----	--------	-------------------------------------

TG:GE0/0/1(U)

.....

```
[SW1]display mac-vlan mac-address all
```

MAC Address	MASK	VLAN	Priority
-------------	------	------	----------

001e-10dd-dd01	ffff-ffff-ffff	10	0
----------------	----------------	----	---

001e-10dd-dd02	ffff-ffff-ffff	10	0
----------------	----------------	----	---

001e-10dd-dd03	ffff-ffff-ffff	10	0
----------------	----------------	----	---

Total MAC VLAN address count: 3



Quiz

1. (Multiple) Which of the following statements about the VLAN technology are incorrect? ()
 - A. The VLAN technology can isolate a large collision domain into several small collision domains.
 - B. The VLAN technology can isolate a large Layer 2 broadcast domain into several small Layer 2 broadcast domains.
 - C. PCs in different VLANs cannot communicate.
 - D. PCs in the same VLAN can communicate at Layer 2.
2. If the PVID of a trunk interface is 5 and the port trunk allow-pass vlan 2 3 command is run on the interface, which VLANs' frames can be transmitted through the trunk interface?



Summary

- This course describes the VLAN technology, including the functions, identification, assignment, data exchange, planning, application, and basic configuration of VLANs.
- The VLAN technology can divide a physical LAN into multiple broadcast domains so that network devices in the same VLAN can directly communicate at Layer 2, while devices in different VLANs cannot.

The background of the image shows silhouettes of several groups of business professionals in a modern office environment. They are standing on a highly reflective floor, and their reflections are clearly visible. The entire scene is overlaid with a semi-transparent blue filter. In the center, the text "Thank You" is written in a large, white, sans-serif font, with the website address "www.huawei.com" in a smaller font directly below it.

Thank You

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